

OMNILABS

RESEARCH



*Professional
Consolidation 1*
Final Report

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Context & Background

This summer, I interned at Omnilabs Research, a medical technology company focused on rehabilitation technology development. For the past few years, Omnilabs has been developing OmniHuman, a remote fine motor experience designed to gameify the tele-rehabilitation for stroke survivors. Its initial prototypes have integrated Apple Vision Pro technology with a game-like interface that challenges users to play games that help develop their fine motor movements. Currently, the company is working on a hand glove that will help facilitate users' finger flexion and extension during these games. Its recent versions utilize pneumatics, incorporating a system of tubes, valves, and bellows to actuate the movement. As Omnilabs aims to manufacture 150 units in the coming months, the team's priority this summer has been finalizing a version for mass production.

Tasks & Responsibilities

Over the course of **eight weeks**, I had three main tasks and reported to the head of engineering.

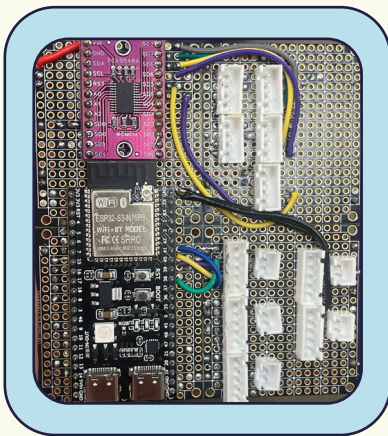
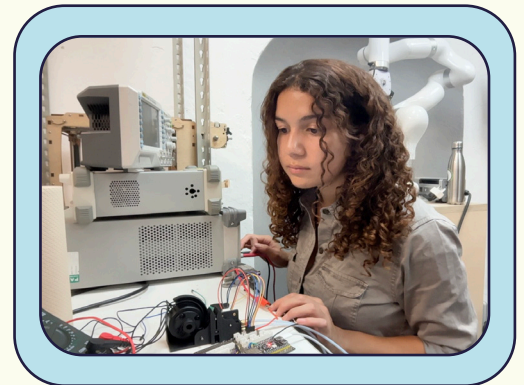
1. Pneumatic System Development

The first version of the glove and core system used pneumatics to actuate hand movement. While the system provides safe and flexible actuation, it also poses significant technical challenges. During the first few weeks of my internship, I worked on fixing the issues that arose with the first few iterations of the pneumatic core. First, we sourced a new power supply and polyurethane tubing. Additionally, I conducted research and testing to determine if future pneumatic cores would require one or five pneumatic pumps to actuate all five fingers. Then, we integrated the new power supply and tubing, and I wrote code for flexion and extension of the hand glove. Finally, we tested the system with the new finger tip extensions.

2. Fully Electric System Design

After meeting with various stakeholders, the team decided to move to an electric system with the finger movement driven by motors, rather than pneumatic pumps, and with magnetic encoders for position tracking. While motors provide less physical flexibility, they are easier to control. Furthermore, electrical systems are more common, making components easier to source.

We opted for 3-phase brushless motors, which are often used in drone technologies. These are driven by motor drivers, and the entire system is powered by a 12V power supply and controlled by an Arduino ESP32-S3.



After we acquired all of the new components, I drew a diagram of the entire system and tested my wiring in the lab, as seen above. After I ensured that everything worked, I soldered components and JST connectors to a protoboard as seen to the left. We conducted further tests with this protoboard and finalized the new glove iterations that are compatible with the motor pulley system.

Finally, I designed a schematic to be sent to the company's third party PCB developers. I connected the board, the multiplexer, the motor drivers, and the connectors. My manager finalized my design and added a Buck converter.

3. Core Design

During my final week, I helped with the core redesign process to incorporate the new electronic components. This included housing the new power supply, the motors, the PCB, the proper tubing, and the magnetic encoders. Although my core design lacks some of the final aesthetic choices, it serves as a framework for future designs.

Skill Development

This summer, I developed my **technical engineering** and **design thinking** skills. Due to the complex nature of this system, I had many opportunities to challenge myself in different directions. Below, I have listed the concrete ways in which my skillset has expanded and developed.

Mechanical Skills

- Developing a pneumatic valve system and integrating it with pumps and cabling
- Determining the proper type of plastic tubing required for various systems and how to successfully route and contain it
- Prototyping hand and finger glove iterations, including developing full flexion and extension of the joints
- Programming a mechanical system with clean and effective Arduino code

Electrical Skills

- Designing an electrical system without any prior documentation
- Researching various motors and their necessary components, including power supply
- Using jumper wires and temporary prototyping to test wire connections and the first pass of Arduino code
- Programming magnetic encoders and determining the correct distance for the magnets to sit from the motor and board to obtain angle readings
- Using a multiplexer to solve I2C address conflicts and connecting multiple magnetic encoders through one Arduino
- Developing, soldering, and testing a protoboard
- Designing a PCB schematic for manufacturing

Design Skills

- Developing housing for both electronic and mechanical component storage in a consumer product
- Building upon my existing Autodesk Fusion skills and challenging my design thinking
- Understanding the necessary design requirements for the patent process
- Learning more about the daily life of an early-stage startup and what it takes to develop, test, deploy, and generate funds for an idea



Reflection & Professional Growth

This summer was a fantastic opportunity for me to grow personally and professionally here in London. Omnilabs operates out of the **Somerset House Makerversity**, a joint community of over 300 founders and creators with dedicated work and making space. Throughout the summer, I met many engineers and creatives and got to learn about the varied work that they do. Furthermore, I had access to all of the wood shops, machine shops, and other resources available.

Going into the project, I was not entirely sure what I would be working on or what type of engineering work I would be doing. As an undergraduate, I majored in Engineering, so I was already familiar with most of the technical components of my work this summer. However, I have never worked directly with pneumatics, and my prior robotics experience was relatively limited, meaning that this was my first time working with drone motors. The technical aspects of my work this summer definitely challenged me to learn more about these applications and to do my own research on stroke rehabilitation as well.

This was my first time working for an early-stage startup. As a result, I learned more about the process of securing funding, applying for grants, joining startup incubators, and how to communicate a new idea to interested stakeholders. Throughout the summer, I felt I was a part of the team and process as Omnilabs developed its first product toward manufacturing.